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Expose `std::atomic_ref` 's object address

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Changelog

- R7: add `constexpr` and add specializations.
- R6: post St. Louis
 - LEWG confirmed `T*` .
 - LEWG picked `address` as the name.
- R5: LEWG September 3rd telecon
 - Update accounting for P3309 and P3323.
 - Update the return type back to `T*` since that is the only design that supports `constexpr` .
 - Updated the name to `get` to align it with the API now returning a pointer (e.g. like `unique_ptr`).
- R4: St. Louis

- Switched to use `uintptr_t`.
- (REVERTED) Update the return type from `const void *` to `address_return_t` which copies the cv-qualifiers of `T`; updated examples, godbolt samples, and returns clause accordingly.
- Include tokyo poll (<https://github.com/cplusplus/papers/issues/1545#issuecomment-2028998320>):
 - POLL:_ Forward "P2835R3: Expose `std::atomic_ref`'s object address" to LWG for C++26 (to be confirmed by an electronic poll).

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Outcome: Consensus in favor

- R3: LEWG mailing list review
 - Per request by one reviewer on the LEWG mailing list, three coauthors of P0019 (Carter H. Edwards, Mark Hoemmen, and Bryce Adelstein Lelbach) joined this proposal to express their approval.
 - Added background section.
 - Added history of why this API was not part of the original paper.
 - Update return type to `const void*`.
 - Rename API to `address()`.
 - Added rationale against `uintptr_t`.
 - Added rationale against `get()` and `data()`.
 - Add new use cases and examples.
 - Removed Discovery Patterns example.
- R2: Preparation for mailing list review
 - Update links to compiler explorer.
 - Update API design rationale.
 - Update `__cpp_lib_atomic_ref` macro.
- R1:
 - Add alternative API designs.
- R0: initial revision (Varna)

Introduction

Applications that need atomic access to an object and want to reason about contention for performance cannot use C++20 `std::atomic_ref`. Some applications may change the object's storage type to `std::atomic`, but `std::atomic_ref`'s raison d'être is that many applications cannot.

This proposal extends `std::atomic_ref` with a member function that returns the object's address. This enables legacy applications that updated their APIs from `volatile int*` to `atomic_ref` to become conforming with the post-C++11 memory model, to recover the optimizations they lost while doing so, while enabling applications still stuck with `volatile*` on their APIs to migrate to use `atomic_ref`.

Before-and-after ("Tony") tables

Before	After
<pre>#include <atomic> #include <cassert> std::atomic<int> data; void fn(std::atomic<int>& ref) { auto* addr = &ref; assert(&data == &ref); } int main() { fn(data); return 0; }</pre>	<pre>#include <atomic> #include <cassert> int data; void fn(std::atomic_ref<int> ref) { int* addr = ref.address(); assert(&data == ref.address()); } int main() { fn(std::atomic_ref{data}); return 0; }</pre>

Motivation

`std::atomic_ref<T>` ensures that all accesses to an existing `T` object are atomic. Unlike with `std::atomic<T>`, for `std::atomic_ref<T>`, the `T` object exists and its lifetime strictly includes all `std::atomic_ref<T>` that refer to it.

Therefore, a `T` object that is used with `std::atomic_ref<T>` could be accessed with both atomic and non-atomic operations during its lifetime (in contrast to `std::atomic<T>`).

However, as long as one or more live `std::atomic_ref` s still reference the `T` object, the object can only be accessed through these `std::atomic_ref` s. That is, non-atomic accesses are not allowed to be concurrent with accesses through `std::atomic_ref` .

Note: this enables implementations of `atomic_ref<T>` to, e.g., copy the value into an `atomic<T>` which is in a different memory location, operate on that, and once the last `atomic_ref` is destroyed, copy the value back (for which implementations must track the address of the original object's location). This proposal does not recommend such an implementation, but it is legal.

The following examples illustrate `atomic_ref` semantics.

This example is well-defined: non-atomic accesses to `data` happen only while there are no live `atomic_ref` s that reference `data` .

```
int data = 13; // Non-Atomic access
{
    assert(data == 13);
    atomic_ref<int> r{data};
    r.store(0); // Atomic access
} // All atomic_refs are destroyed here

// No atomic_refs are live:
assert(data == 0);
data = 42;
```

This example exhibits undefined behavior: a non-atomic access to `data` happens while there is still a live `atomic_ref` that references `data` .

```
int data = 13;           // Non-Atomic access
atomic_ref<int> r{data}; // atomic_ref live
data = 42;               // UB: object accessed during lifetime of atomic_re
```

The implementation of APIs using `std::atomic<T>*` may obtain the object address without breaking API changes.

```
void api_atomic(std::atomic<int>* ptr) {
    // can obtain address of underlying object:
    uintptr_t address = reinterpret_cast<uintptr_t>(ptr);
}
```

The implementation of APIs using `std::atomic_ref<T>` may not:

```

void api_atomic_ref(atomic_ref<int> ref) {
    // ...cannot obtain the address of underlying object
    // if constexpr(sizeof(ref) == sizeof(uintptr_t)) {
    //     uintptr_t address = reinterpret_cast<uintptr_t>(&ref); // UB
    // }
}

```

This proposal extends the `atomic_ref` API with a member function `address()` to obtain the underlying object's address.

```

void api_atomic_ref_this_paper(atomic_ref<int> ref) {
    // With this proposal, one can get the underlying object's address
    uintptr_t address = reinterpret_cast<uintptr_t>(ref.address());
}

```

Intent of `atomic_ref` proposal

The paper that introduced `atomic_ref` in C++ 20 is P0019R8 (<https://wg21.link/P0019R8>). The authors discussed this use case and decided the application should track `&data` themselves. However, APIs evolve as usage patterns emerge: SG1 reviewed this paper to address this oversight, and forwarded it with *unanimous consent*. Multiple authors of the original `atomic_ref` paper are co-authors of this paper.

Use cases

This proposal enables legacy APIs that are still using `volatile*` to signal concurrency, due to their implementations needing the object's address internally (see Motivation), to finally migrate to the C++11 Memory Model.

Some of the reasons why these APIs need the object address are covered in this section. Others, like "C Foreign Function Interface (FFI)," are not currently covered in this document.

Atomic access to elements of a data structure

Applications that want to perform atomic access to the elements of a data structure need to make the data structure's element type `atomic`,

```

| std::array<std::atomic<int>, N> array;

```

and use pointers to `atomic` objects to access the elements.

```
int fetch_add_idx(std::atomic<int>* base, size_t i, int value) {
    return base[i].fetch_add(value);
}
```

When the array is provided externally, e.g., from a third-party C API, it is typically an array of `T`, not an array of `atomic<T>`.

```
extern int array[N];
```

Before `atomic_ref` was introduced in C++20, it was common practice for applications to create APIs that drop the "atomicity" semantics. Such applications would use `volatile` (with nonstandard meaning) to express their intent.

```
int fetch_add_idx(/* not atomic */ int volatile* base, size_t i, int value) {
    return std::atomic_ref{base[i]}.fetch_add(value);
}
```

However, it is not possible for applications written in Standard C++ to encode "atomicity" semantics as part of their API without a way to extract the underlying's object address. This proposal provides that way:

```
int fetch_add_idx(std::atomic_ref<int> base, size_t i, int value) {
    auto* p = base.address();
    return std::atomic_ref{*(p+i)}.fetch_add(value);
}
```

This matches the original intended use case of `std::atomic_ref<T>`, namely accessing the elements of an existing array of `T` atomically.

In fact, for the particular case of contiguous data structures, like the array above, `std::atomic_ref` was specifically designed to be a proxy reference type for `mdspan` accessors. The `atomic_accessor` class in P2689 (<https://wg21.link/p2689>) has an `access(T* p, size_t i)` member function that returns `std::atomic_ref{*(p+i)}`. The `mdspan` proposal P0009 (<https://wg21.link/p0009>) used `atomic_ref` and its corresponding accessor as justification for `mdspan` permitting custom accessors. (See e.g., the "Why custom accessors?" section of P0009.)

Use atomic accesses only when required

One of the most valuable features `atomic_ref` provides is being able to mix atomic and non-atomic accesses to an object as required.

In the following example, multiple threads concurrently access memory, and then signal when they are done by incrementing a counter. The last thread must perform extra work that accesses the memory, but since there are no other threads concurrently accessing the memory, these accesses do not need to be atomic:

```
void thread(atomic_ref<int>* data, atomic_ref<int> counter, int nthreads) {
    data->fetch_add(42, memory_order_relaxed);
    int* d = data->address(); // get raw pointer to data
    data->~atomic_ref(); // destroy this thread's data's atomic ref
    int pos = counter.fetch_add(1); // signal data and destruction
    if (pos != (nthreads - 1)) return;
    // last thread: does not need to use atomic accesses
    int last_data = *d; // non-atomic access
    // ...
}
```

With `.address()`, these APIs can use `atomic_ref` in the signature, expressing that the data will be concurrently accessed, and then obtain a raw pointer for non-atomic accesses if allowed.

Without `.address()`, these APIs would need to either:

- use `int*` in the type signature, and then construct an `atomic_ref` internally,
- use `atomic_ref` for all accesses.

Contention-aware data-structures

Contention-aware data-structures rely on the object's address to tell different objects apart. The addresses are then used to, e.g., index into global lock tables.

Feedback during LEWG mailing review suggested that examples that were too advanced were unapproachable. R2 of the paper provided a fully working implementation of one such example (<https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2024/p2835r2.html#Discovery-Patterns>) in compiler-explorer, demonstrating a 1.25x performance improvement from this API. Similar algorithms are part of, e.g., C++ standard library implementations (e.g., here

(https://github.com/NVIDIA/cccl/blob/b7d4228ab7268ed928984cd61096079bd671d25d/libcudacxx/include/cuda/std/detail/libcxx/include/__cuda/barrier.h#L231-L250)).

Design

The name and return type should prevent accidental misuse that could result from accessing the object through the address while there are still live `atomic_ref` that reference it.

Furthermore, this proposal interacts with the following proposed extensions:

- P3323 (<https://wg21.link/p3323>) (cv-qualified types in `atomic` and `atomic_ref`): SG1 forwarded it to LEWG in St. Louis, which will most likely forward it to LWG after electronic poll. Therefore, the solutions explored here need to handle `atomic_ref<cv-qual T>`.
- P3309 (<https://wg21.link/p3309>) (`constexpr atomic` and `atomic_ref`): SG1 forwarded it to LEWG in St. Louis, which will most likely forward it to LWG after electronic poll. Therefore, the solutions explored here need to support `constexpr`.

We considered the following options.

- Return type:
 1. `T*` (see P3323 (<https://wg21.link/p3323>)):
 - **PRO:** supports all use cases and inflight proposals.
 - **CON:** potential accidental access to the referenced object while there are still `atomic_ref` objects live, resulting in undefined behavior.
 2. `uintptr_t`:
 - **PRO:** prevents accidental use of the return value to access the object, requiring a `reinterpret_cast` to do so.
 - **CON:**
 - `uintptr_t` is an optional type, and therefore this API would need to be optional.
 - `reinterpret_cast` is currently not supported in `constexpr`, and therefore, converting the value back to a pointer to access the object would not be allowed.
 3. `address_return_t / cv-qual void*`: copy cv-qualifiers of `T` to a `void*`
 - **PRO:** prevents accidental use of the return value to access the object, requiring a `static_cast` from/to cv-qualified `void*` to do so.
 - **CON:**
 - Casting from `void*` is currently not supported in `constexpr`, and therefore, converting the value back to a pointer to access the object would not be allowed.

```
using address_return_t
= conditional_t<is_const_v<T> && is_volatile_v<T>, void const volatile*,
conditional_t<is_const_v<T>, void const*,
conditional_t<is_volatile_v<T>, void volatile*, void*>>>;
```

- Name:
 - `get` :
 - Requirement: `T*` return type.
 - **PRO**: well-aligned with Standard Library APIs. All APIs with a `get` member function return a pointer, and types like `unique_ptr` already have preconditions on when that pointer can be used (e.g. `unique_ptr` returns `nullptr` in some cases), object lifetimes, etc.
 - **CON**: may suggest to some that the pointer can be accessed.
 - `address` :
 - **PRO**: suggests that the address can be used without suggesting that the object may be accessed.
 - **CON**: unfamiliar name.
 - `data` :
 - Requirement: `T*` return type.
 - **CON**: types with a `data` member function always have a `size` member function and the two functions together indicate that the type represents a contiguous range. A `data` member function would thus incorrectly suggest that `atomic_ref` also represents a contiguous range. Also, `mdspan` deliberately rejected the name `data` in favor of `data_handle`, because `mdspan` might not actually view a contiguous range and because `data_handle_type` might not necessarily be `element_type*`.
 - `ref` :
 - **CON**: incorrectly suggest that the return value may be used to access the object, just like `std::reference_wrapper`.

This design proposes using `T*` as the return type because we deem `constexpr` support more valuable than protecting against some accidental undefined behavior, given that writing programs that accidentally exhibit undefined behavior is quite easy due to data-races.

This design proposes using `address` since `get` implies that the object may be directly accessed.

A prototype implementation with some tests is available here ([godbolt.org](https://gcc.gnu.org/bugzilla/show_bug.cgi?id=81668))

(https://gcc.gnu.org/bugzilla/show_bug.cgi?id=81668).

Impact on implementations

This proposal does not impact any implementation we are aware of. We surveyed libc++, libstdc++, Microsoft STL, and libcu++.

Wording

NOTE: If P3323 is accepted, the return type `T*` below is to be kept as `T*` to propagate cv-qualifiers (i.e., do not replace with `value_type`).

Add the following to [atomics.ref.generic.general] (<http://eel.is/c++draft/atomics.ref.generic.general>).

```
namespace std {
    template<class T> struct atomic_ref {
        private:
            T* ptr; // exposition only
        public:
            // ...
            constexpr T* address() const noexcept;
            // ...
    };
}
```

Add the following to [atomic.ref.ops] (<http://eel.is/c++draft/atomics.ref.ops>):

```
constexpr T* address() const noexcept;
```

Returns: `ptr`.

Add the following to [atomics.ref.int] (<https://eel.is/c++draft/atomics.ref.int>):

```
namespace std {
    template<> struct atomic_ref<integral-type> {
        ...
        public:
        ...
        constexpr integral-type* address() const noexcept;
    };
}
```

Add the following to [atomics.ref.float] (<https://eel.is/c++draft/atomics.ref.float>):

```

namespace std {
    template<> struct atomic_ref<floating-point-type> {
        ...
        public:
        ...
        constexpr floating-point-type* address() const noexcept;
    };
}

```

Add the following to [atomics.ref.pointer] (<https://eel.is/c++draft/atomics.ref.pointer>):

```

namespace std {
    template<> struct atomic_ref<pointer-type> {
        ...
        public:
        ...
        constexpr pointer-type* address() const noexcept;
    };
}

```

Update `__cpp_lib_atomic_ref` version macro in `<version>` synopsis [version.syn] (<https://eel.is/c++draft/version.syn#2>) to the C++ version this feature is introduced in:

```

#define __cpp_lib_atomic_ref 201806____L // freestanding, also in <atomic>

```